

Deep learning approach for short-term entry passenger flow forecasting in urban rail transit stations

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ARTICLE INFO

Keywords:

Urban rail transit
Short-term passenger flow forecasting
Deep learning
Hybrid model

ABSTRACT

In short-term passenger flow forecasting for urban rail transit, Automatic Fare Collection (AFC) data serves as a crucial carrier reflecting the characteristics of passenger flow variations. Existing short-term passenger flow prediction methods primarily rely on historical time-series data from individual stations, yet similar passenger flow trends may exist across different stations. To comprehensively capture the passenger flow characteristics of various station types and enhance the interpretability of predictions, this paper proposes a CEEMDAN-TCAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise and Temporal Convolutional Attention Network) method based on two-step clustering. First, a two-step fuzzy k -means (FkM) approach is employed to perform spatial feature extraction based on land use patterns and passenger flow characteristics around metro stations, thereby providing a more generalized dataset for station-level predictions. Subsequently, CEEMDAN is applied to deeply extract multi-scale temporal features of passenger flow, improving the prediction accuracy of station entry passenger volumes by addressing the automatic adjustment of non-stationary time-series data. Next, an encoder-decoder framework is adopted for station-level passenger flow prediction, and the model's rationality is validated using attention score heatmaps at different step lengths and partial autocorrelation coefficients. To verify the effectiveness of the proposed method and model, extensive experiments are conducted using historical passenger flow data from Suzhou Metro stations. The results demonstrate that the proposed model achieves 1.7 %–36.9 % higher prediction accuracy compared to other classical models while reducing average training time by over 55.8 %.

1. Introduction

Urban metro systems have become a vital mode of transportation for residents due to their advantages in safety, speed, affordability, and comfort. However, the rapid increase in metro passenger numbers has led to a series of prominent issues, such as longer queues and waiting times at entry/exit gates, overcrowding inside trains, and more frequent emergency incidents. Short-term passenger flow forecasting involves predicting near-future passenger flow trends based on historical data patterns. Accurate passenger flow prediction not only helps operators and relevant authorities adjust station crowd management strategies and optimize train scheduling but also enables early warnings for potential emergencies, allowing timely passenger dispersal and peak-hour travel guidance (LIU et al., 2019). Nevertheless, most existing methods lack interpretable results and probabilistic statistical information, which further limits their potential for improving prediction accuracy (Wang et al., 2024).

In the field of big data applications for intelligent rail transit, the AFC system stores massive historical passenger flow data, recording travel timestamps, entry/exit volumes, and passenger flow directions. These data can reveal temporal patterns of passenger flow and provide critical value for short-term passenger flow prediction at urban rail transit stations. Many scholars have classified metro stations based on AFC historical data and Point of Interest (POI) data, establishing short-term passenger flow forecasting frameworks to study how fares affect urban travel demand (Huang et al., 2018; Li et al., 2017; Vlahogianni et al., 2014; Wang et al., 2018). However, existing methods predominantly rely on simplistic clustering approaches (e.g., k -means) for station categorization, which fail to effectively group stations with homogeneous characteristics—ultimately compromising the accuracy of short-term passenger flow predictions.

Eliminating random noise in passenger flow data and improving result interpretability represent critical challenges for composite models in urban rail transit short-term passenger flow prediction. Traditional

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methods, including Empirical Mode Decomposition (EMD), Ensemble EMD (EEMD), and Variational Mode Decomposition (VMD), are prone to mode mixing, which compromises decomposition quality and limits their applicability across scenarios. In contrast, the CEEMDAN demonstrates superior noise robustness and computational efficiency when handling complex time series characterized by non-stationarity and nonlinearity (Chen and Wei, 2024; Li et al., 2025; Pan et al., 2025; Zhang et al., 2025). Meanwhile, Temporal Convolutional Networks (TCNs) have gained widespread adoption in sequential modeling tasks, including time series forecasting (Bai et al., 2018). However, when processing lengthy input sequences without delivering interpretable results, TCN architectures may require excessive convolutional layers, a constraint that escalates computational overhead while obscuring decision logic.

To address the limitations in prediction interpretability and insufficient probabilistic statistical information, this study constructs a CEEMDAN-TCAN model based on two-step clustering that integrates stations with similar passenger flow characteristics. For more accurate categorization of stations with homogeneous patterns, a two-step FkM module is introduced to obtain optimized parent-child clustering results, thereby establishing a robust data foundation for passenger flow forecasting. Furthermore, the proposed CEEMDAN-TCAN model enhances prediction accuracy and interpretability by: (i) extracting temporal features from passenger flow data through CEEMDAN, and (ii) incorporating a sparse attention mechanism to effectively identify crucial historical patterns.

The main contributions of this work can be summarized as follows:

- (1) To comprehensively capture passenger flow characteristics of similar station types, this paper proposes a two-step FkM. Initially, parent station clusters are generated through fuzzy clustering based on station attributes and passenger flow metrics. Subsequently, child station clusters are derived by applying k-means clustering to each parent cluster, ultimately screening the most representative station dataset.
- (2) To address the limitations in prediction accuracy and model interpretability, we propose a CEEMDAN-TCAN framework for passenger flow forecasting. This model processes modal-decomposed passenger flow sequences from categorized stations and performs prediction under a sparse attention mechanism, which effectively enhances the model's noise resistance and long-sequence processing capability, thereby achieving superior prediction accuracy and interpretability.
- (3) Extensive experiments were conducted on a week-long dataset encompassing 5 metro lines and 169 stations, with results demonstrating the superior performance of the proposed method over baseline approaches. Furthermore, comprehensive ablation studies were performed to validate: (i) the impact of historical time steps on prediction accuracy, and (ii) the operational rationality of the attention mechanism.

The structure of this paper is organized as follows: Section 2 presents the foundational work, including cluster analysis, modal decomposition techniques, and relevant forecasting methodologies. Section 3 details the proposed research framework, encompassing data preprocessing procedures, spatiotemporal data mining approaches, and the novel station-level short-term passenger flow prediction model. Section 4 provides a comprehensive case study analysis, which involves: (i) anomaly detection and data processing, (ii) spatiotemporal feature extraction from rail transit data, and (iii) model training, comparative analysis, and performance evaluation. Finally, Section 5 concludes with key findings and suggests directions for future research.

2. Related work

Existing short-term passenger flow forecasting methods can be

primarily categorized into three classes: traditional mathematical statistics (Tang et al., 2019), machine learning, and deep learning prediction models. Mathematical statistics-based approaches include ARIMA (Ma et al., 2020), Kalman filtering (Zhou et al., 2019), and logistic regression prediction (Xu and Jiang, 2020). While these methods feature fixed model complexity and fewer parameters, they exhibit limited capacity for comprehensive time series fitting and struggle to capture nonlinear characteristics in the data (Gu et al., 2020; Roos et al., 2017; Smith and Demetsky, 1997; Yao et al., 2020).

Benefiting from the continuous advancement of data mining techniques, machine learning has been adopted by researchers for short-term passenger flow forecasting. Models such as Decision Trees (Ding et al., 2016; Xu et al., 2020), Random Forests (Zarei et al., 2013), Multilayer Perceptrons (Halawa et al., 2016), and Support Vector Machines (Li et al., 2021; Sun et al., 2015; XU et al., 2022) have demonstrated enhanced prediction accuracy in short-term passenger flow forecasting. While machine learning-based prediction models can effectively capture uncertainty factors and nonlinear characteristics in passenger flow fluctuations - providing innovative technical solutions to overcome the limitations of traditional mathematical models in handling randomness - they remain constrained by shallow feature representation. These models fail to adequately account for the impact of metro network structures on passenger flow, making it difficult to capture the complex spatiotemporal characteristics of traffic flow data.

With the continuous advancement of neural networks, deep learning neural networks featuring more hidden layers and enhanced capabilities for learning higher-dimensional data features and complex data relationships have emerged as a research focus. While Recurrent Neural Networks (RNNs) are typically employed to model temporal dependencies in data, their inherent architectural limitations can lead to gradient vanishing and explosion issues when processing long sequences, making model training challenging (Hopfe et al., 2024). Prediction models based on the Long Short-Term Memory (LSTM) framework can effectively address these issues and achieve relatively accurate passenger flow forecasting results (Baghbani et al., 2023; Yang et al., 2019). Additionally, numerous innovative feedforward models have been applied to time series modeling, with notable examples including TCNs (Bai et al., 2018) and the self-attention-based Transformer architecture (Vaswani et al., 2023), which enable modeling of long-term dependencies in multivariate time series data. The TCN architecture primarily comprises causal convolutions, dilated convolutions, and residual connections, achieving an expanded receptive field while mitigating unstable gradient issues and improving training speed. However, TCNs remain a black-box architecture that neither learns positional dependency relationships within sequences nor extracts internal correlations from inputs, consequently failing to provide interpretable results. Recent years have seen researchers applying Transformers to passenger flow prediction with promising outcomes (Yue and Ma, 2023). Nevertheless, Transformers suffer from high computational complexity, insufficient sensitivity to local temporal patterns, and a requirement for extensive training data to prevent overfitting.

Existing passenger flow prediction models have increasingly adopted composite approaches, achieving improved forecasting accuracy. However, they still exhibit limitations in spatiotemporal feature extraction and result interpretability. Huang et al. (2023) employed CEEMDAN to decompose raw passenger flow data into Intrinsic Mode Functions (IMFs) and residuals, subsequently applying a Seq2Seq model for multi-step short-term forecasting. Similarly, Zeng et al. (2022) developed a hybrid model combining CEEMDAN with LSTM for short-term passenger flow prediction. While these models demonstrate capability in learning spatiotemporal features from passenger flow data, they overlook the dynamic evolution of inter-station spatiotemporal correlations, and fail to simultaneously address both global flow characteristics and model interpretability in a unified learning framework.

In this work, we propose a two-step clustered CEEMDAN-TCAN

architecture. On one hand, inspired by TCN, we employ dilated causal convolution to capture input causal relationships. On the other hand, by incorporating a sparse attention mechanism, the model effectively extracts internal correlations while learning position-aware dependencies, thereby significantly reducing the computational complexity of long-sequence forecasting.

3. Methodology

The methodology section consists of four subsections. Section 3.1 discusses the data preprocessing methods. Section 3.2 proposes a two-step FkM clustering approach. Section 3.3 covers modal decomposition theory, detailing the computational process of the CEEMDAN and providing comparative analyses. Section 3.4 introduces the TCN's advantages and limitations, incorporates a sparse attention mechanism, and presents the TCAN model framework. Finally, the integrated CEEMDAN-TCAN model framework is summarized.

3.1. Data preprocessing

The raw unprocessed data are generally unsuitable for direct analytical applications. Although AFC data effectively capture passenger travel patterns, errors may occur during data collection, transmission, processing, and storage. To enhance data quality, rigorous data cleansing is essential. Given the relatively low error probability of AFC data in practice, this study focuses on identifying and eliminating erroneous and redundant records rather than performing data imputation or correction.

Current approaches for processing anomalous data can be broadly categorized into: statistical-based, distance-based (Knorr and Ng, 1998), density-based, and clustering-based methods. The Local Outlier Factor (LOF) algorithm represents a classic density-based anomaly detection approach, which identifies anomalies by analyzing the local density deviation of samples relative to their neighbors - a lower density indicates higher probability of being an outlier. Existing research demonstrates that anomalous data in rail transit passenger flows typically occurs within specific temporal ranges. While global anomaly detection methods may yield counterproductive results, density-based approaches can simultaneously examine both local and global perspectives, thereby more effectively identifying passenger flow anomalies across different dates and months. Consequently, this study employs the density-based LOF algorithm to perform anomaly detection on AFC data.

First, in dataset S , let $dist_k(p)$ be the distance from data point p to its k -th nearest neighbor. Then, the set of data points whose distance from p does not exceed $dist_k(p)$ is defined as the k -distance neighborhood of p , denoted as $N_k(p) = \{q | dist(p, q) \leq dist_k(p), q \in S(x \neq p)\}$, where $dist(p, q)$ represents the distance between data objects p and q . The k -reachable distance between data objects p and o is denoted as $rdist_k(p, o) = \max\{dist_k(o), dist(p, o)\}$. The local reachability density of p , denoted as $l_k(p)$, is defined as the reciprocal of the average reachable distance from points in neighborhood $N_k(p)$ to p , as shown in Eq. (1).

$$l_k(p) = 1 / \left(\frac{\sum_{o \in N_k(p)} rdist_k(p, o)}{|N_k(p)|} \right) \quad (1)$$

The $l_k(p)$ reflects the distribution of the data within $N_k(p)$, the lower the $l_k(p)$ of the data, the higher the probability of anomaly. The ratio of the local reachability density of the data within the neighborhood $N_k(p)$ of data p to $l_k(p)$ is the LOF of p , as in Eq. (2).

$$LOF_k(p) = \frac{\sum_{o \in N_k(p)} \frac{l_k(o)}{l_k(p)}}{|N_k(p)|} \quad (2)$$

$LOF_k(p)$ reflects the anomaly degree of point p . When $LOF_k(p) < 1$, it indicates that the density of p is higher than that of its k -nearest

neighbors $N_k(p)$, suggesting p is a normal value. Conversely, when $LOF_k(p) > 1$, p is potentially an outlier. The LOF algorithm primarily identifies anomalies based on $LOF_k(p)$ through the following procedure: First, set the parameter k (number of neighbors) and a threshold λ_{LOF} for the local outlier factor. Then, compare each data point's LOF against λ_{LOF} , eliminating those points where $LOF > \lambda_{LOF}$ from the dataset.

3.2. A two-step fuzzy K-mean algorithm

Existing research encompasses various clustering methods. Fuzzy clustering is particularly suitable for data with fuzzy or overlapping characteristics, adapting to more complex distributions. It not only generates cluster partitions but also provides membership degrees of each data point to different clusters, facilitating subsequent analysis. However, this approach suffers from high computational complexity on large datasets and may compromise interpretability. In contrast, k -means clustering has gained widespread application due to its fast convergence and simple principles, effectively compensating for Fuzzy clustering's interpretability limitations. For metro station clustering, two conventional methods are typically employed to define station attributes (Gan et al., 2020): The first approach utilizes temporal distribution characteristics of passenger flow, such as the ratio of hourly passenger volume to daily volume or peak-hour volume. While this indicator captures daily variation trends of passenger flow, it somewhat neglects the absolute magnitude of passenger volume, potentially resulting in stations within the same cluster having similar variation patterns but significantly different flow volumes. The second method directly employs hourly passenger volume to characterize station attributes. Although this effectively captures flow magnitude variations, it partially overlooks fluctuation patterns, which may lead to stations with comparable flow volumes but divergent variation trends being grouped together.

To comprehensively capture both passenger flow variation trends and volume magnitudes during the clustering process, this paper proposes a two-step FkM clustering methodology. The first step performs fuzzy clustering on stations using temporal distribution indicators of passenger flow to obtain parent-class clusters. The second step conducts k -means clustering within each parent class based on hourly passenger flow volumes, generating child-class station groupings. Building short-term prediction models using aggregated passenger flow data from all stations within child classes effectively expands the training dataset, thereby enhancing prediction performance. The implementation involves: (i) calculating temporal distribution indicators according to Eq. (3) while statistically analyzing surrounding land-use characteristics; (ii) executing fuzzy clustering through the passenger flow indicator matrix combined with station-area land-use attributes as formulated in Eq. (4); with (iii) optimal cluster number determination via Eq. (5).

$$FI_{sj} = \frac{F_{sj}}{\sum_j F_{sj}} \quad (3)$$

$$J_m(U, V) = \min_{U, V} \left\{ \sum_{i=1}^k \sum_{j=1}^n U_{ij}^m x_j - v_{i2} \right\} \quad (4)$$

$$s = \frac{b - a}{\max(a, b)} \quad (5)$$

where, FI_{sj} represents the passenger flow index of the s station in the j hour, and F_{sj} represents the passenger flow of the s station in the j hour; U represents membership matrix, V represents cluster center, and x_j represents input index data. a indicates the average distance between a sample and other samples in the cluster, and b indicates the average distance between a sample and other samples in the cluster.

Finally, for the results of each parent class, the k -means clustering is carried out for the stations under each parent class by using F_{sj} passenger flow matrix, and the loss function is calculated as shown in Eq. (6). The

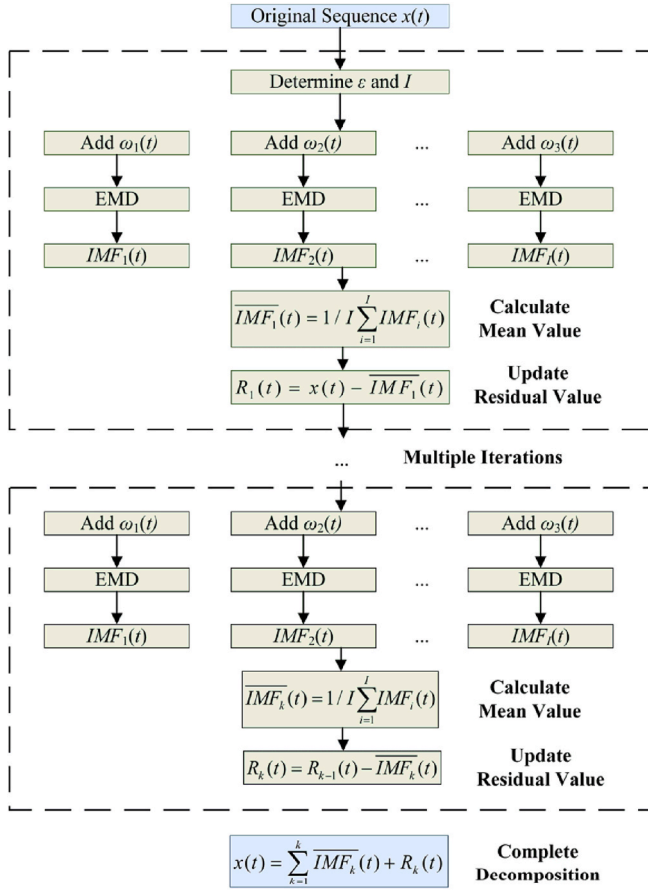


Fig. 1. The CEEMDAN architecture.

subclass stations obtained by the two-step FkM clustering algorithm have the same passenger flow trend and passenger flow size.

$$Loss = \sum_{i=1}^k \sum_{j=1}^{n_k} \min_{X_{ij} \in X, C_i \in C} (\|X_{ij} - C_i\|^2) \quad (6)$$

where, X_{ij} is the station sample, C_i is the clustering center, n_k is the station data in each category, and k is the number of categories.

3.3. A complete ensemble empirical mode decomposition with adaptive noise

EMD is a decomposition technique for nonlinear and non-stationary signals that separates data into several IMFs and a residual sequence based on their time scales; however, EMD has notable limitations: firstly, due to uneven distribution of extreme points during decomposition, the resulting IMFs may suffer from mode mixing where a single component contains multiple time-scale characteristics, and secondly, although EMD requires multiple iterative computations, the lack of well-defined termination criteria means different stopping conditions can yield inconsistent decomposition results.

To address the limitations of traditional EMD, EEMD incorporates white noise of equal amplitude to improve the uneven distribution of extreme points in the data, ensuring the uniqueness of the decomposed modal components, where the final modal decomposition results require multiple averaging to achieve the effect of mutual cancellation of white noise. However, EEMD exhibits poor completeness in reconstructing the original signal by summing the decomposed components, potentially generating multiple low-frequency IMFs with minimal amplitude that contribute little to signal analysis, ultimately resulting in low computational efficiency.

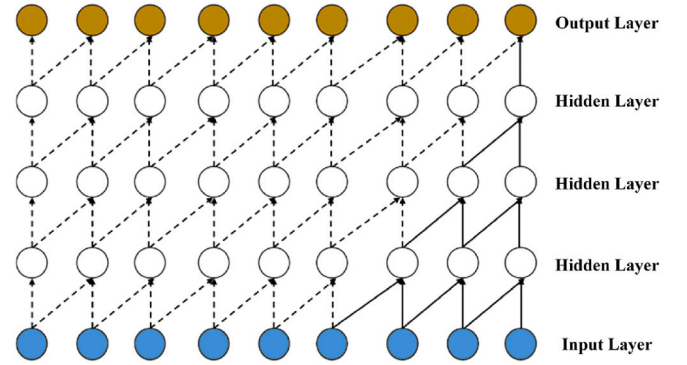


Fig. 2. A causal convolutional structure.

Additionally, the CEEMDAN algorithm is an adaptive data pre-processing method (Torres et al., 2011) that decomposes data into a set of IMFs and a Res arranged from high to low frequencies, capturing the characteristics of the original data across different time scales. Compared to EEMD, CEEMDAN's IMFs can more accurately reconstruct the original signal while eliminating certain noise-dominated IMFs (see Fig. 1). The CEEMDAN decomposition process begins by adding adaptive Gaussian white noise $\omega_i(t)$ to the original urban rail transit passenger flow time series data $x(t)$, obtaining $x_i(t) = x(t) + \beta_0 \omega_i(t)$. After CEEMDAN decomposition, the first-order $\overline{IMF}_1(t) = 1/I \sum_{i=1}^I IMF_i^1(t)$ and first-order residual $R_1(t) = x(t) - \overline{IMF}_1(t)$ are obtained. Using $R_1(t)$ as the new sequence, the decomposition continues for $R_1(t) + \beta_1 E_1[\omega_i(t)]$ to yield $\overline{IMF}_2(t) = 1/I(E_1\{R_1(t) + \beta_1 E_1[\omega_i(t)]\})$ and $R_2(t) = R_1(t) - \overline{IMF}_2(t)$. This iterative process continues until the $k+1$ st-order $\overline{IMF}_{k+1}(t) = 1/I(E_k\{R_k(t) + \beta_k E_k[\omega_i(t)]\})$ and k th-order residual $R_k(t) = R_{k-1}(t) - \overline{IMF}_k(t)$ are obtained. The CEEMDAN decomposition terminates when $R_k(t)$ becomes monotonic; otherwise, the decomposition continues. Ultimately, $x(t)$ is decomposed into: $x(t) = \sum_{k=1}^k \overline{IMF}_k(t) + R_k(t)$.

3.4. A temporal convolutional attention-based network

TCN is an improved neural network method for processing time series based on CNN. Compared to LSTM and GRU, TCN uses one-dimensional convolutional neural networks for time series prediction, extracting features within specific time steps through convolutional kernels, demonstrating superior performance for time series forecasting. Relative to traditional CNNs, it exhibits further improved feedback and convergence speeds while enhancing the network's learning capacity. Figs. 2 and 3 illustrate TCN's causal convolution and dilated convolution modules, which enhance the ability to capture historical information. Additionally, to prevent information loss caused by excessive layers, TCN employs residual modules to replace single convolutional layers, enabling cross-layer information transfer in the neural network as shown in Fig. 4.

As shown in Fig. 2, the input layer is composed of a set of sequences of $x_{1:T} = x_1, \dots, x_T$. Both the hidden layer and the output layer are obtained by synthesizing the input sequences and sequence order of the previous layer, and finally output a set of $\hat{y}_{1:T} = \hat{y}_1, \dots, \hat{y}_T$ sequences. In the case of too many hidden layers, a large amount of input data can only generate a single output result, which will lead to the problems of vanishing gradient, training complexity and poor fitting. The use of expansion convolution allows receptive field to expand, which can solve part of the problem of causal convolution, as shown in Fig. 3. In the expansion convolution can be interval sampling, the sampling rate is determined by receptive field, $dilation = 1$ in the input layer, that is, for each point into the next layer of computation; hidden layer of $dilation = 2$, is every two points into the next layer of computation; the higher the number of layers, the larger the $dilation$, the larger the interval of its

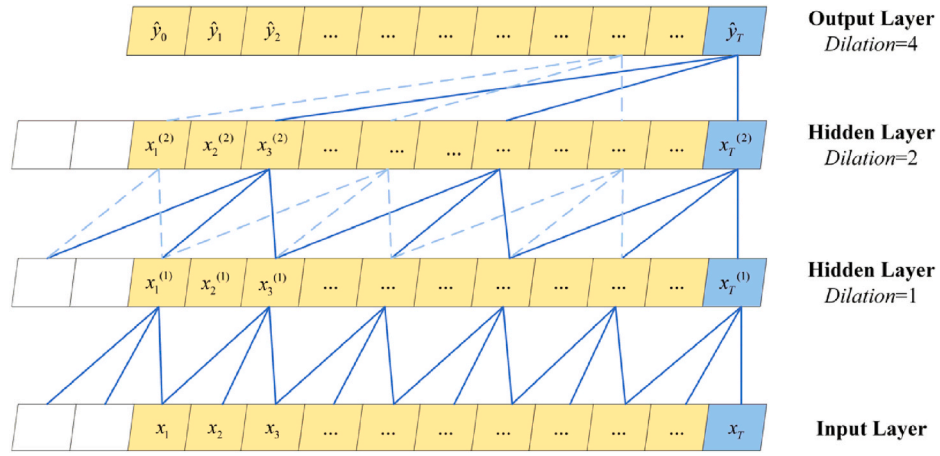


Fig. 3. A dilated convolution structure, kernel size = 3.

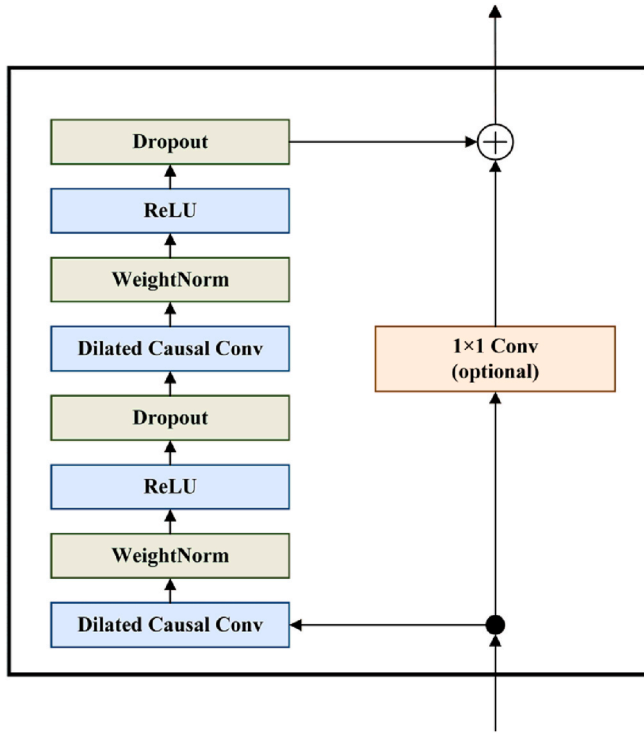
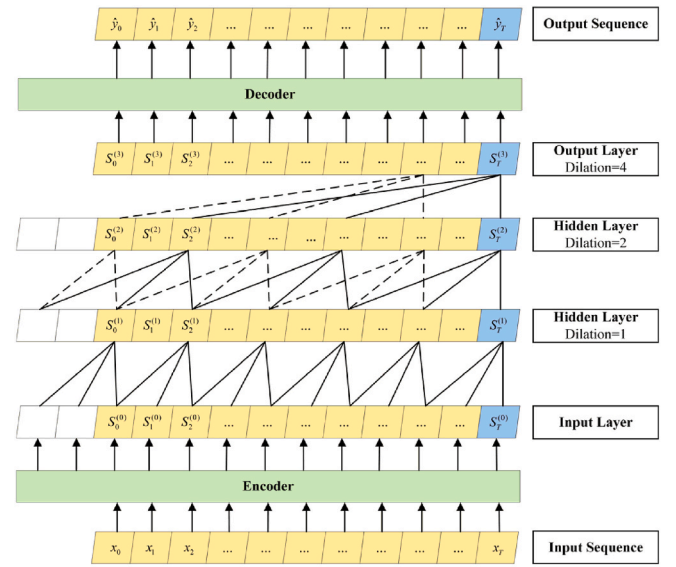


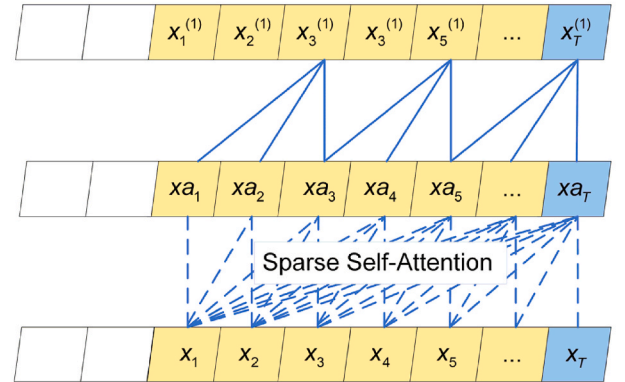
Fig. 4. A residual module.

entry into the next layer. The expansion convolution effective window increases rapidly with the increase of the number of layers, the convolutional network can get the maximum receptive field under fewer layers.

The whole TCN framework consists of the above three modules, but TCN also suffers from the problems of poor interpretability and slow training due to too many convolutional layers. The self-attention mechanism can establish global dependencies between inputs and outputs, effectively circumventing recursive structures (Lin et al., 2022) as shown in Eq. (7). Q denotes the matrix that packs a set of attention functions on a query, K and V denote the matrices of keys and values, and d_k denotes the dimension of the input keys. The Transformer is a powerful sequence model, but its required time and memory grow quadratically with sequence length. Child et al. (2019) proposed sparse factorization of attention matrices, reducing computational complexity to $O(n\sqrt{n})$ while saving memory usage, enabling the network to model



(a) Model overall



(b) TCAN block

Fig. 5. Overall architecture and interlayer transformation diagram. (a) Overall architecture including Input, Hidden and Output layers; (b) Inter-layer transformation of TCAN.

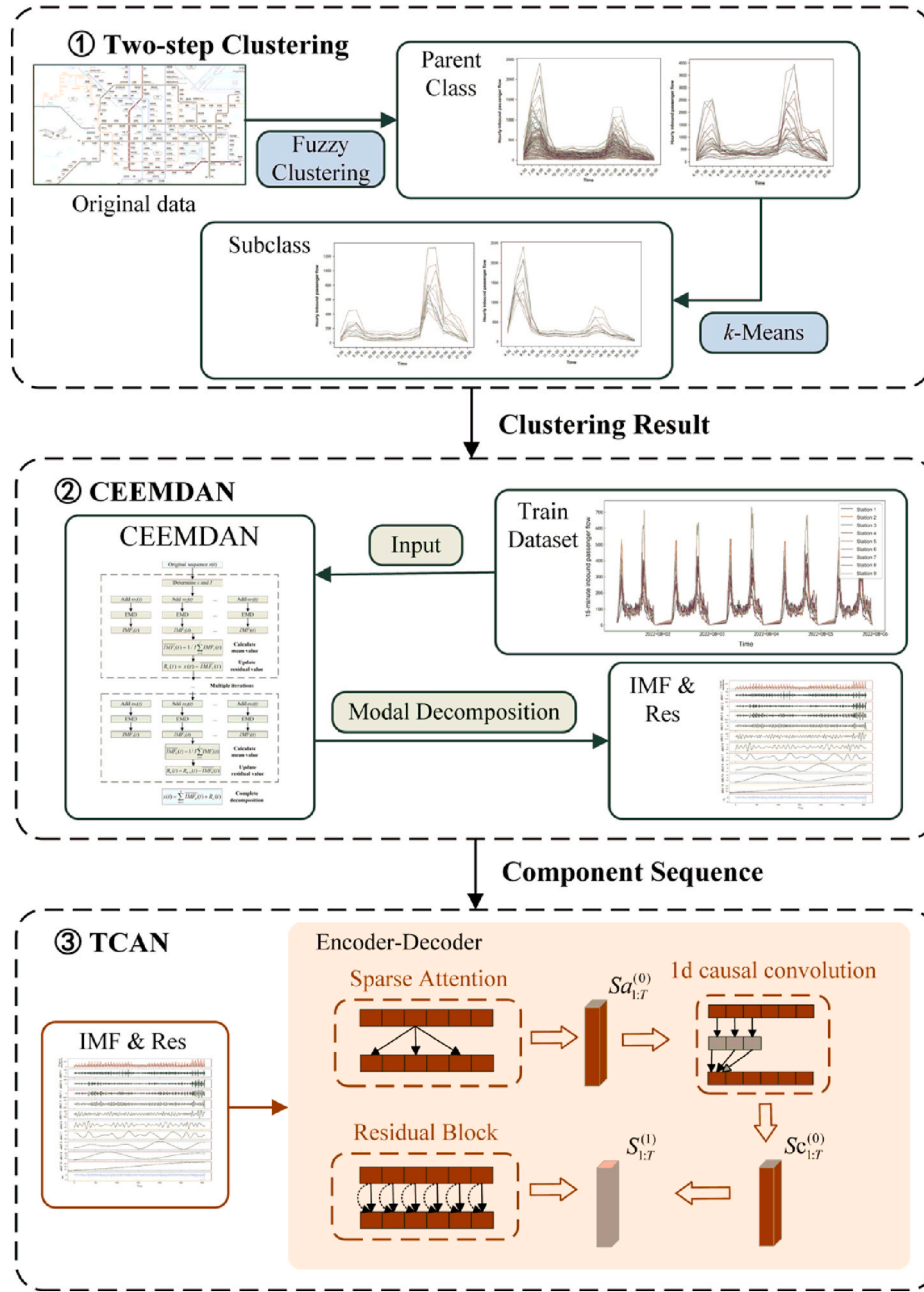


Fig. 6. TCAN model framework.

sequences with extensive timesteps. Compared to self-attention mechanisms, sparse attention tends to generate fewer attention scores to better capture crucial historical steps while disregarding irrelevant historical intervals to prevent their impact on prediction accuracy. Therefore, this study employs sparse attention mechanisms for modeling passenger flow sequences at metro stations.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

In Fig. 5, the TCAN adopts a structure similar to TCN, comprising an encoder and a decoder. At the beginning of the model, the encoder maps the input sequence $x_1 : T = x_1, x_2, x_3, \dots, x_T$ to a continuous sequence $S_{1:T}^{(0)} = \text{Encoder}(x_{1:T})$ at the input layer, where T denotes the sequence length and 0 represents the 0-th layer. Subsequently, dilated causal convolutional hidden layers with different kernel sizes are applied. After the final hidden layer, the decoder generates an output sequence $\hat{y}_{1:T} =$

$\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T$. Within the TCAN block, the input sequence $S_{1:T}^{(0)}$ is processed through a sparse attention mechanism to become $S_{1:T}^{(0)}$, representing an intermediate variable containing temporal step information. Building upon $S_{1:T}^{(0)}$, we apply a 1D causal convolution to obtain the convolved output sequence $S_{1:T}^{(0)}$. Finally, $S_{1:T}^{(1)}$ merges the intermediate variable $S_{1:T}^{(0)}$ with the residual connection provided by $S_{1:T}^{(0)}$, completing an inter-layer transformation process.

It can be recognized that the encoder-decoder framework can handle variable-length input and output sequences, though its limitations lie in the inability to comprehensively characterize full-sequence information and the potential replacement or diminished influence of initially input sequence data by subsequent sequences, where the introduced sparse attention mechanism effectively compensates for these shortcomings. The TCAN employs TCN's hierarchical convolutional structure to extract temporal dependencies while utilizing sparse attention to focus on

Table 1
Selected AFC card data.

Card type	Date of arrival	Stop time	Inbound station number	Outbound date	Outbound time	Outbound station number
100	20220801	19	469	20220801	219	470
100	20220731	233041	151	20220801	41	470
151	20220731	232502	149	20220801	516	265
255	20220731	230420	249	20220801	116	270
59	20220731	210753	147	20220801	15248	147
59	20220801	3811	529	20220801	3847	529
59	20220731	234647	467	20220801	2823	467

critical time steps. Notably, TCAN's sparse attention layer supports an expanded receptive field without requiring deeper architectures and simultaneously enhances the interpretability of prediction results (Hao et al., 2023).

3.5. Station-level short-term passenger flow prediction model

- (1) Two-step fuzzy k-means based station clustering. A two-step clustering approach is proposed to classify stations according to passenger flow characteristics and surrounding land-use features. First, a fuzzy clustering algorithm generates parent station clusters, followed by K-means clustering to obtain child station clusters, resulting in a classified dataset of stations with similar passenger flow variation trends and volume scales.
- (2) Modal decomposition-based temporal feature extraction. The CEEMDAN method is employed to decompose passenger entry flow data, effectively reducing flow volatility. The decomposed IMFs and Res components are then used as original input data for prediction, enabling temporal feature mining of rail transit time-series data to enhance model prediction precision and accuracy.
- (3) CEEMDAN-TCAN prediction model. By integrating a sparse attention mechanism into the TCN framework, a two-step clustering-based CEEMDAN-TCAN model is developed for station-level short-term passenger flow prediction within an Encoder-Decoder architecture, which better captures critical historical steps while addressing issues of weak interpretability and low training efficiency. Since stations within the same category exhibit similar passenger flow patterns, the model can be trained using all station data belonging to the same category, with the model architecture illustrated in Fig. 6.

4. Case study

4.1. Dataset

The data used in this study were obtained from the gate swipe records of Suzhou Metro's AFC system, covering a one-week period from August 1 to August 7, 2022, including 5 weekdays and 2 non-working days, with the research area encompassing 5 lines and 169 stations generating over 1 million data entries in tabular format containing fields such as card type, entry date, entry time, entry/exit station codes, exit date, exit time and exit station codes as shown in Table 1.

4.2. Data preprocessing

Through preliminary analysis of AFC data, the following key data cleaning steps were determined: first, removing staff tickets and converting time formats; then, filtering out abnormal data (e.g., records where entry time is later than exit time, exits occurring outside operating hours, cross-day trips, and trips with excessively short or long durations) and eliminating duplicate records; finally, extracting passenger flow volumes at different time granularities for each station.

Given the date dataset $L = \{1, 2, \dots, L, \dots\}$, the time unit length Δt , and the time period set $T = \{1, 2, \dots, t, \dots\}$, where time period t represents the time range $[(t-1)\Delta t, t\Delta t]$, the process begins by setting k and

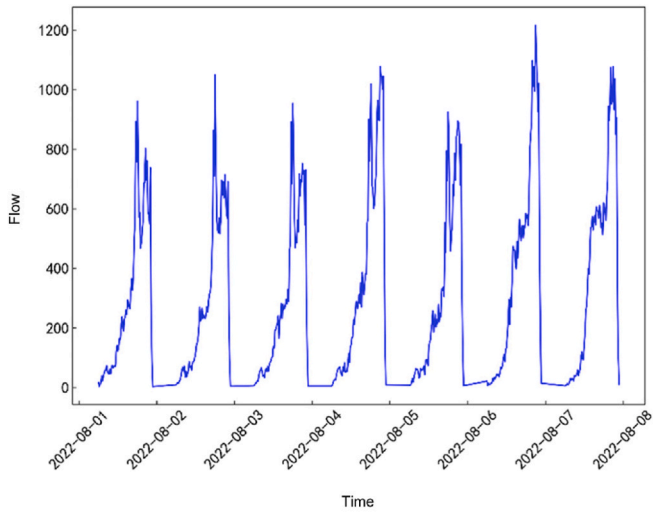
establishing the traffic data sample set. A data sample set $S_i = [s_i(1), s_i(1), \dots, s_i(L)]$ is created for the daily time-divided entry passenger flow at station i , where $s_i(L)$ denotes the data sample set of station i within time interval L , composed of entry passenger flow for each time period. Next, LOF for each sample is calculated and its threshold is determined. Following Equations (1) and (2), the LOF within each time interval of S_i is obtained, with Euclidean distance serving as the distance metric as specified in Eq. (8). The Local Outlier Factor is typically used to identify data anomalies, where $LOF > 1$ indicates the presence of anomalous samples in the dataset, while $LOF \leq 1$ signify normal data. Due to varying dataset distributions, probabilistic statistical methods are commonly employed to determine the LOF threshold. In statistical analysis, box plots provide an intuitive representation of dataset distributions without imposing significant constraints on data objects or types; thus, this study utilizes box plots. By setting the inner limit $limit_{LOF} = Q_3 + 1.5$ for S_i , the threshold for the local outlier factor is defined as $\lambda_{LOF} = \max\{1, limit_{LOF}\}$. Finally, anomalous passenger flow data is detected, with data points where $LOF > \lambda_{LOF}$ identified as anomalies. Upon identifying routine passenger flow data, corrections are applied based on the mean values of passenger flow data from the preceding and subsequent time periods.

$$d(s_i(x), s_i(y)) = \sqrt{\sum_{t=1}^T (s_i(x) - s_i(y))^2} \quad (8)$$

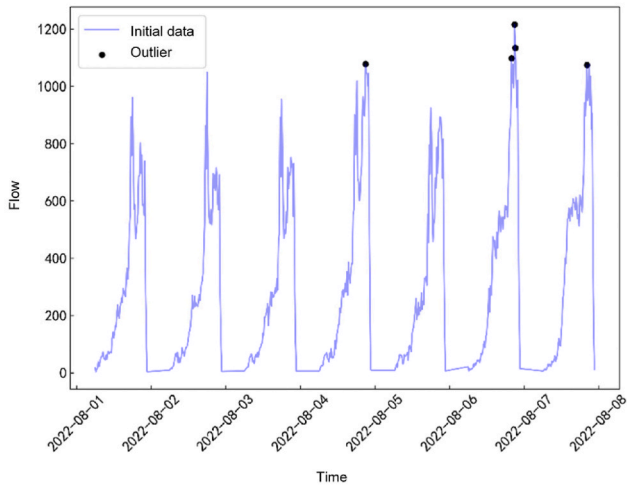
This study employs the local outlier factor for anomaly detection and data cleansing, implemented in Python using the Sklearn, Numpy, and Pandas packages. To enhance the efficiency, entry passenger flow data undergoes normalization. Taking Suzhou Dongfangzhimen Station as a case study, the initial data distribution comprises 15-min interval entry passenger flow records in Fig. 7(a). Parameter k selection critically influences anomaly detection outcomes, with experimental results identifying optimal model parameters $k = 20$ that are subsequently set as thresholds. Fig. 7(b) presents the anomaly detection results, where outliers are replaced with mean values to maintain data continuity, with processed results shown in Fig. 7(c). Analysis reveals five anomalous points occurring exclusively during peak hours over a weekly observation period. While AFC swipe data generally maintains high quality with no anomalies during off-peak periods, unavoidable mis-swiping behaviors during peak hours generate sporadic abnormal records. Comparative analysis between Fig. 7(b)–(c) demonstrates that the combined local outlier factor detection and mean imputation method preserves overall flow trends while selectively replacing only distinct outliers.

4.3. Temporal correlation analysis

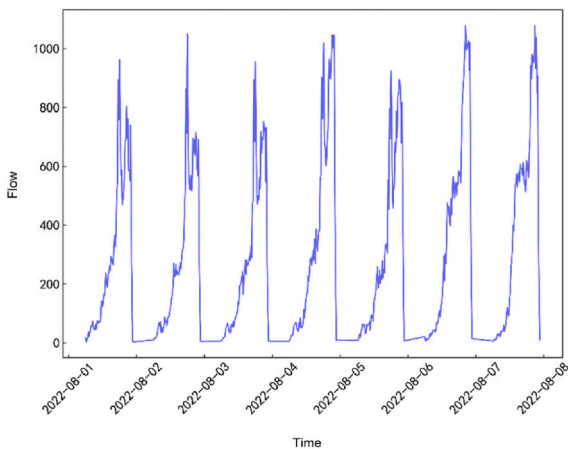
The grey relational analysis method is a data analysis approach that can be used to examine the degree of correlation between different date-based time series data, with its fundamental principle being to determine the strength of association between variables based on the closeness of their relationship. From a temporal perspective, the entry passenger flow in rail transit systems exhibits certain regularities, typically demonstrating periodic and similar variation trends. This paper takes Dongfangzhimen Station as an example to analyze the correlation between weekdays and non-weekdays, as illustrated in Fig. 8.



(a) Inbound traffic flow

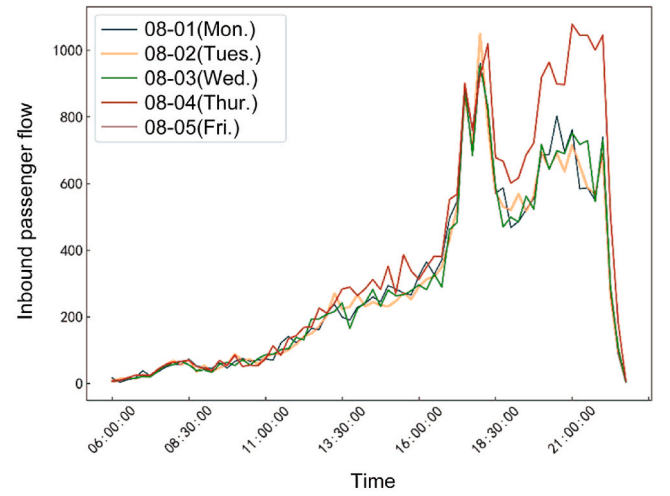


(b) Outlier detection

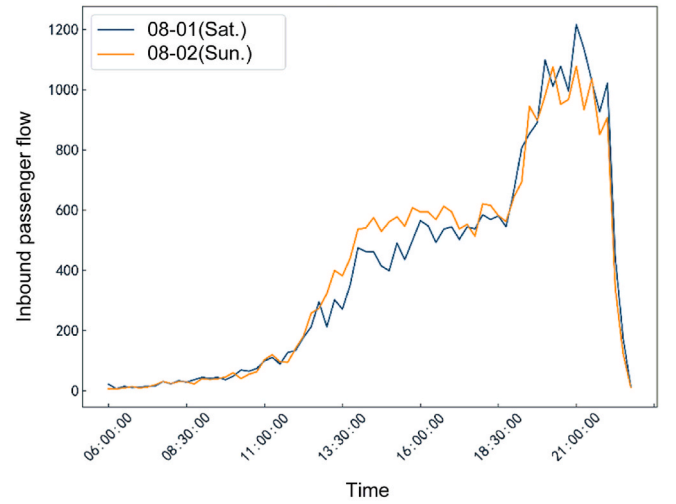


(c) outlier replacement

Fig. 7. Data preprocessing case study.



(a) weekdays



(a) weekends

Fig. 8. The temporal distribution of entry passenger flow on weekdays and non-weekdays.

Table 2

The correlation degree aggregation between workdays and non-workdays.

Correlation degree	Mon.	Tues.	Wed.	Thur.	Fri.	Sat.	Sun.
Mon.	1.00	0.90	0.92	0.81	0.84	0.80	0.71
Tues.	0.89	1.00	0.89	0.77	0.80	0.76	0.69
Wed.	0.92	0.90	1.00	0.82	0.85	0.80	0.72
Thur.	0.83	0.81	0.84	1.00	0.89	0.84	0.76
Fri.	0.85	0.82	0.86	0.88	1.00	0.84	0.76
Sat.	0.79	0.77	0.79	0.81	0.83	1.00	0.83
Sun.	0.71	0.71	0.72	0.74	0.76	0.84	1.00

In Fig. 8(a), the variation trends of entry passenger flow on weekdays are generally similar, showing a steady increase from 00:00 to 16:00, entering the evening peak phase at 16:00 which lasts until 18:30, followed by a second evening peak phase at 21:00. This passenger flow pattern aligns well with actual geospatial attributes as the Dongfangzhimen Station is located in downtown Suzhou, being a commercial-type station that exhibits dense passenger flow during evening peaks. As shown in Fig. 8(b), the variation trends of entry passenger flow on non-weekdays are also generally similar, with the evening peak phase occurring between 18:30 and 21:00. The travel purposes on non-

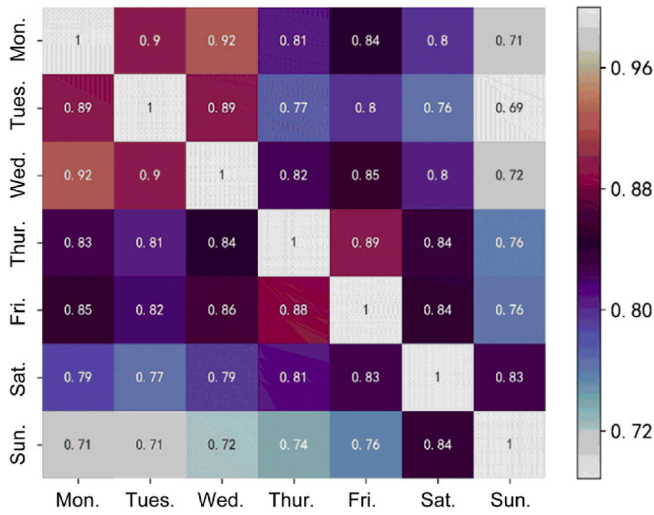
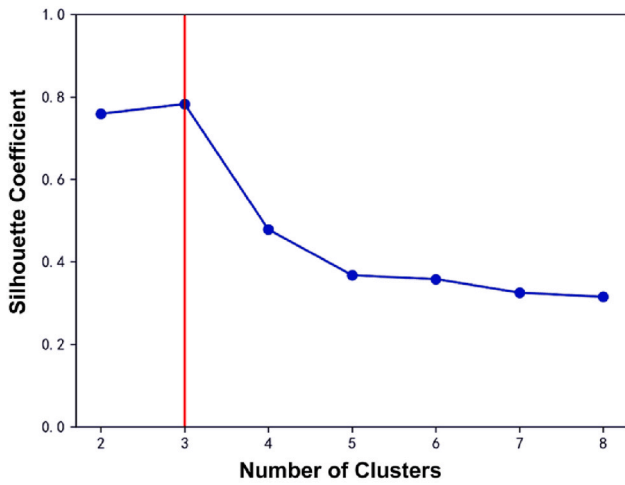
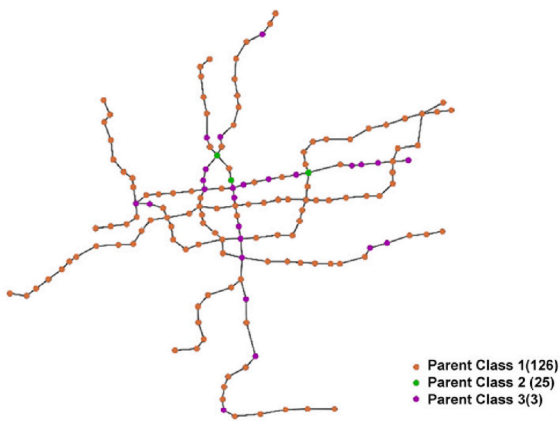


Fig. 9. The correlation heatmaps between weekdays and non-weekdays.

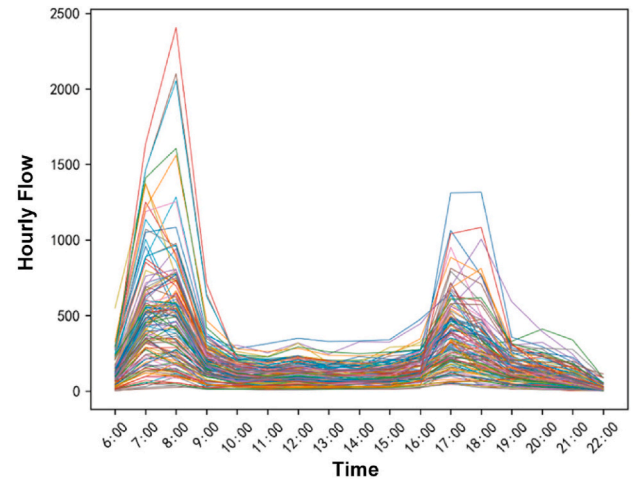


(a) Parent clustering results

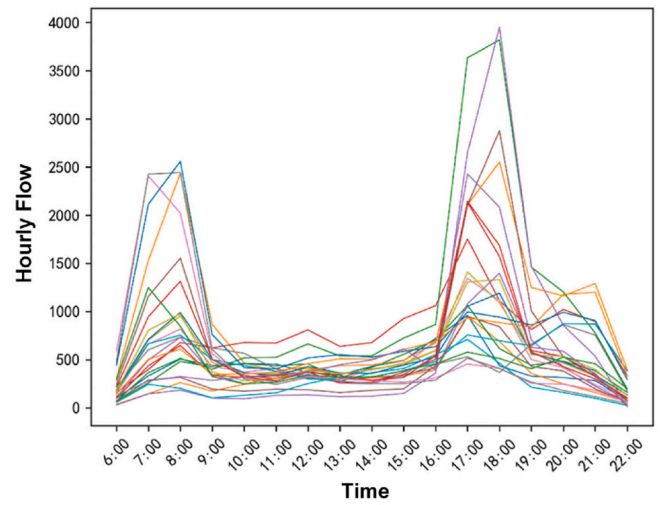


(b) Spatial distribution of different parent stations

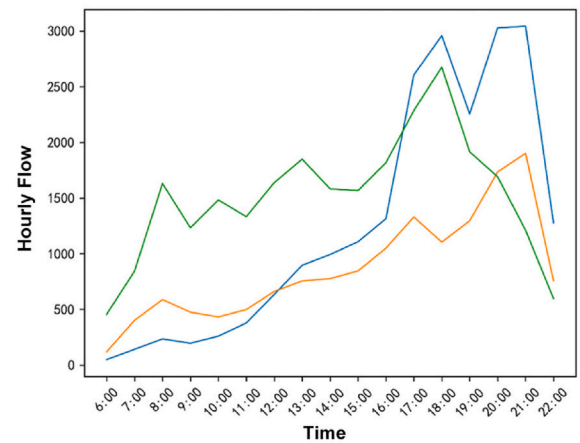
Fig. 10. Clustering analysis of metro entry swipe card data.



(a) Parent-class 1



(b) Parent-class 2



(c) Parent-class 3

Fig. 11. The temporal distribution of entry passenger flow across three parent-class station types.

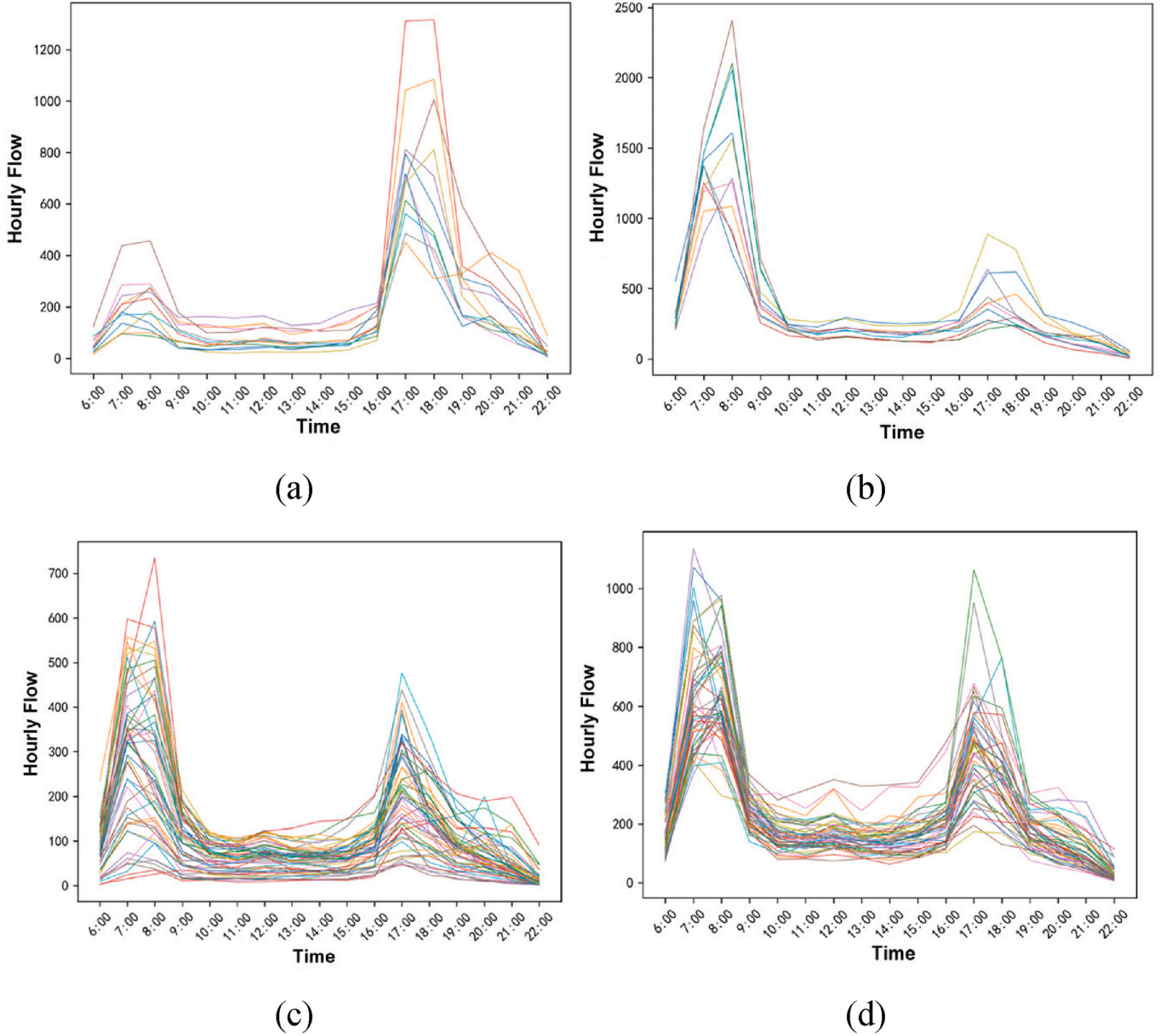


Fig. 12. The temporal distribution of inbound passenger flow across the four subcategories within Parent Category 1, (a) Parent class 1- Subclass 1, (b) Parent class 1- Subclass 2, (c) Parent class 1- Subclass 3, (d) Parent class 1- Subclass 4.

weekdays are primarily leisure and entertainment, which differ significantly from weekday travel patterns.

According to Table 2 and Fig. 9, the correlation between weekdays is above 0.8, while the average speed correlation between weekdays is 0.78; however, the correlation between weekdays and non-weekdays is significantly lower, indicating distinct variation trends in rail transit entry passenger flow between these periods. Weekdays exhibit greater station passenger flow fluctuations with a bimodal pattern compared to non-weekdays, demonstrating the necessity of distinguishing between these day types. Due to data volume limitations, this study exclusively focuses on weekday data.

4.4. Spatial feature mining based on two-step fuzzy K-mean

The dataset used in this study consists of five consecutive workdays (August 1–5, 2022) of metro entry swipe card data, where Fig. 10(a) presents the parent-class results from the first-step fuzzy clustering, and since metro stations typically cluster into 3 to 8 groups, the Silhouette

Coefficient begins to decline and stabilize when the number of clusters exceeds 3, with 3 clusters representing a clear inflection point, as evidenced by Fig. 10(b) which displays the spatial distribution of these three parent-class stations.

Stations within the same cluster exhibit identical passenger flow trends, and this paper provides a concise analysis of the passenger flow characteristics for these three parent-class station categories. Parent-class 1 stations demonstrate higher morning peak entry passenger flows compared to evening peaks, with relatively stable levels during other periods, showing distinct dual-peak characteristics as illustrated in Fig. 11(a). Predominantly located in suburban areas, Parent-class 2 stations exhibit greater evening peak entry passenger flows than morning peaks in Fig. 11(b), with most stations situated near urban centers, classifying them as typical “work-residence hybrid stations.” Parent-class 3 stations display prominent single-peak patterns without distinct morning rush periods, primarily located in central business districts, constituting typical “work-oriented stations” as shown in Fig. 11(c).

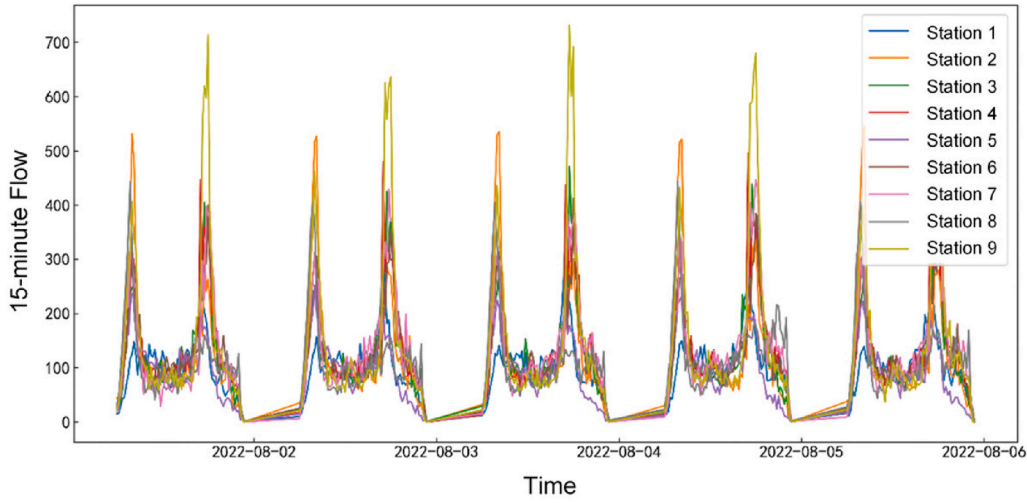


Fig. 13. Training data trend.

Stations within the same parent category exhibit similar spatial distribution characteristics and passenger flow trends, yet demonstrate significant variations in entry passenger flow volumes; therefore, after completing the initial fuzzy clustering step, a secondary k -means clustering becomes essential, though constrained by the limited number of stations in parent category 3, this study exclusively performs secondary clustering on stations from parent categories 1 and 2, taking parent category 1 as an example, k -means reclusters the 126 stations into 4 subcategories whose temporal distributions of entry passenger flow are illustrated in Fig. 12, ultimately, the two-step clustering results will serve as the input dataset for the CEEMDAN-TCAN model.

4.5. Short-term passenger flow prediction based on CEEMDAN-TCAN model

4.5.1. Experimental setup and evaluation metrics

During the model training process, the parameters are updated and the CEEMDAN-TCAN is trained using a gradient descent-based back-propagation algorithm, with the model's minimum objective loss function as shown in Eq. (9). Employing various gradient descent optimization strategies to address parameter update issues in back-propagation algorithms is a current research focus, with *Adam* being one of the most straightforward and popular algorithms (Kingma and Ba, 2017), which this paper adopts as the algorithmic basis for gradient descent.

$$L(\theta) = \frac{1}{2N} \sum_{h=1}^N (\hat{y}_h - y_h) \quad (9)$$

where, $\hat{y}_h$ is the predicted value, y_h is the true value, and N is the size of the training set.

The parameters of CEEMDAN-TCAN are iteratively updated through the algorithm. Before model training, the obtained entry passenger flow data of the same station type is divided into two parts: 70 % training set and 30 % validation set, with the validation set data completely excluded from the model parameter optimization process. As the CEEMDAN-TCAN model outputs probabilistic predictions, it provides corresponding confidence intervals ρ under different states.

This paper selects Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) as performance evaluation metrics, where MAE represents the average error between predicted and actual values, MSE evaluates the magnitude of errors between predicted and actual values, and MAPE describes the relative error, with the calculation formulas shown in Eq. (10).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{\hat{y}_i} \right| \times 100\%$$

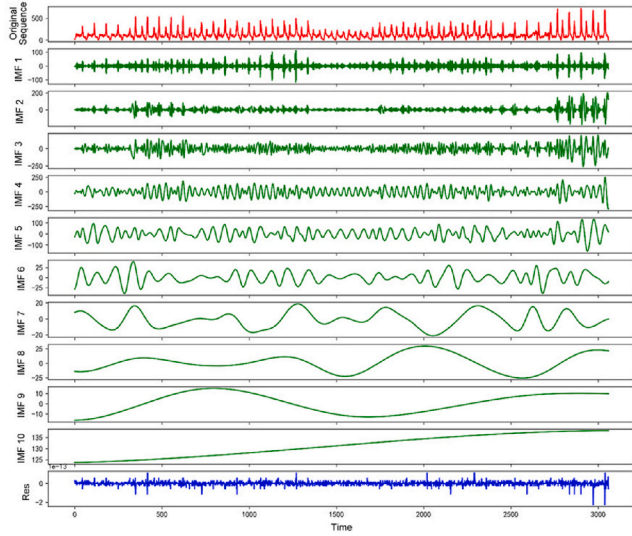
where $\hat{y}_i$ represents the predicted value, y_i denotes the actual value, and n indicates the sample size.

The CEEMDAN-TCAN model was compared with other mainstream deep learning forecasting models, including DeepAR, N-BEATS, LS-Transformer, TCNN-3, TCNN-4, and TCAN, using the aforementioned metrics, where the parameters of these models were adjusted based on existing research. The hyperparameters of all models were uniformly selected through Bayesian optimization for hyperparameter search, with the maximum number of epochs set to 200 and the maximum number of iterations set to 20 for all models. The experiments were conducted on a 64-bit PC platform equipped with an i7-11800 @ 2.30 GHz CPU, 16 GB RAM, and a GeForce RTX 3050Ti GPU.

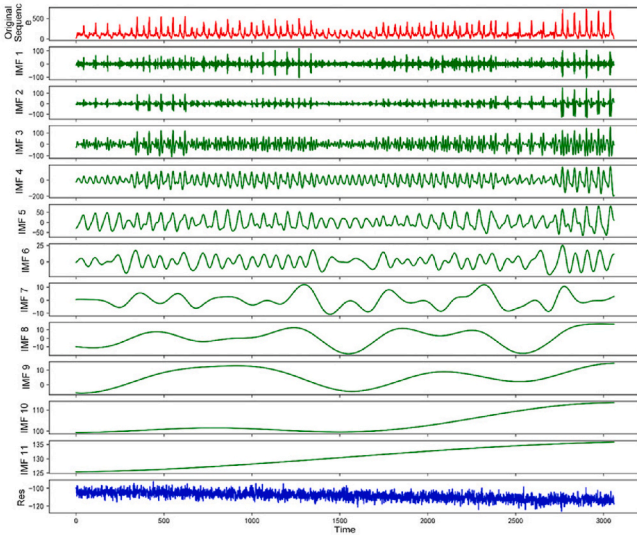
4.5.2. Mode decomposition

In terms of temporal granularity selection, a 15-min interval is widely recognized by most scholars as a reasonable time granularity range, and this paper validates using swipe data at 15-min intervals. According to the two-step clustering results in Section 4.4, the station dataset of Parent Category 2 - Child Category 1 was selected for case validation, with trend graphs of 9 stations shown in Fig. 13. Although peak values differ across stations, their overall passenger flow trends exhibit similarity. Therefore, to address the challenge of limited metro swipe data availability, datasets from stations of the same type were concatenated into an integrated dataset for model training. This concatenation expanded the dataset from 340 records per individual station to 3060 records, sufficiently meeting the requirements for model training.

According to the requirements of the proposed model, the network input consists of the entry passenger flow data from the past 14 consecutive time intervals along with the current time t 's entry passenger flow data $X = \{F_{t-14}, F_{t-13}, \dots, F_t\}$, while the network output is the entry passenger flow data $Y = \{F_{t+1}, \dots, F_{t+4}, F_{t+5}\}$ for the next 5 time intervals, with each interval spanning 15 min. To enhance training efficiency and address dimensional influences, the dataset was normalized



(a) CEEMDAN



(b) EEMD

Fig. 14. Modal decomposition result.

to constrain the entry passenger flow values between 0 and 1. The augmented dataset was then decomposed using both CEEMDAN and EEMD methods, with the modal decomposition results shown in Fig. 14. The EEMD method decomposed the augmented entry passenger flow sequence into 12 components (11 IMFs and 1 Res), whereas CEEMDAN separated the sequence into 10 IMFs and 1 Res. The time component sequences from both methods were individually reconstructed and compared with the original entry passenger flow sequence to obtain reconstruction errors. CEEMDAN eliminated certain noisy IMF components, achieving a reconstruction error on the order of 10–15 (effectively zero), while the EEMD algorithm yielded larger reconstruction errors with excessive low-frequency IMF components. Consequently, this study adopts CEEMDAN for modal decomposition of station-level short-term entry passenger flow sequences.

4.5.3. Model training and analysis

The CEEMDAN-decomposed component sequences are individually fed into the TCAN model for training, and the output results are

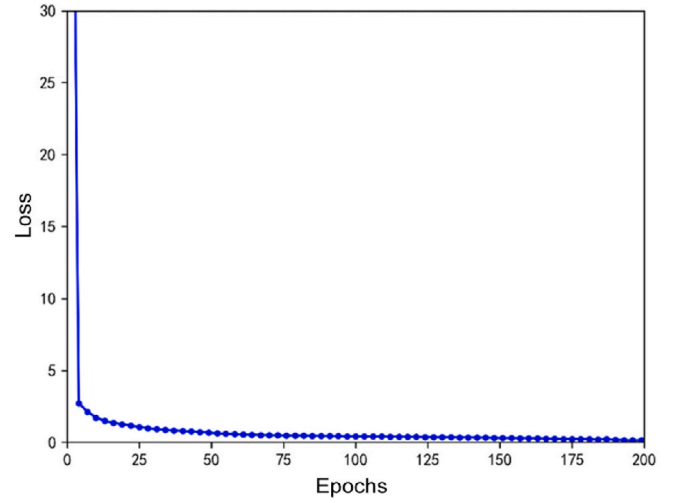


Fig. 15. The decline of the training loss function.

superimposed to obtain the final predicted values. Figs. 15 and 16 illustrate the model's training process and prediction results, demonstrating that the proposed model achieves strong fitting performance for passenger flow forecasting.

To further demonstrate the predictive performance of the model, it is compared with DeepAR, N-BEATS, LS-Transformer, TCNN-3, TCNN-4, and TCAN models as shown in Table 3. Using a 15-min time granularity, the differences between deep learning models are not significant due to limited volatility impact. Under these conditions, CEEMDAN-TCAN achieves the smallest error. Among probabilistic forecasting models, DeepAR performs relatively well with a MAPE of 9.98 %, demonstrating the effectiveness of autoregressive recurrent neural network structures for probabilistic forecasting. The difference between pure feedforward architecture-based models and Transformer-based models is not significant, indicating that multi-layer fully connected modules and sparse attention mechanisms are less efficient in time series forecasting and struggle to capture meaningful features. TCAN achieves higher accuracy than both TCNN-4 and TCNN-3, proving the effectiveness of sparse attention mechanisms in improving accuracy and expanding receptive fields without adding more convolutional layers. Both TCAN and TCNN-4 can cover all values in the input sequence with their receptive fields, but TCAN uses fewer convolutional layers than TCNN-4. Compared with TCAN, CEEMDAN-TCAN reduces MAE by 13.61 %, RMSE by 14.56 %, and MAPE by 4.1 %, demonstrating that CEEMDAN integration can effectively extract time series inputs and reduce the impact of data noise on prediction results.

As shown in Table 3, the CEEMDAN-TCAN model achieves optimal accuracy across different confidence probabilities while the TCNN-4 model also demonstrates excellent predictive performance, and although increasing convolutional layers can enhance the model's forecasting capability to some extent, excessively adding hidden layer depth may significantly compromise computational efficiency; therefore, this study compares the training speed between CEEMDAN-TCAN and TCNN-4, both of which can expand the receptive field to cover the entire input sequence length, and experimental results from three runs under identical conditions reveal that CEEMDAN-TCAN reduces average training time by over 55.8 % compared to TCNN-4 as illustrated in Fig. 17, which can be readily explained by CEEMDAN-TCAN's fewer temporal convolutional layers, reduced trainable hyperparameters, and implementation of sparse attention mechanism to eliminate unnecessary historical step computations.

4.5.4. Model validation

Comparative model analysis demonstrates that the CEEMDAN-TCAN-based prediction model achieves superior performance in

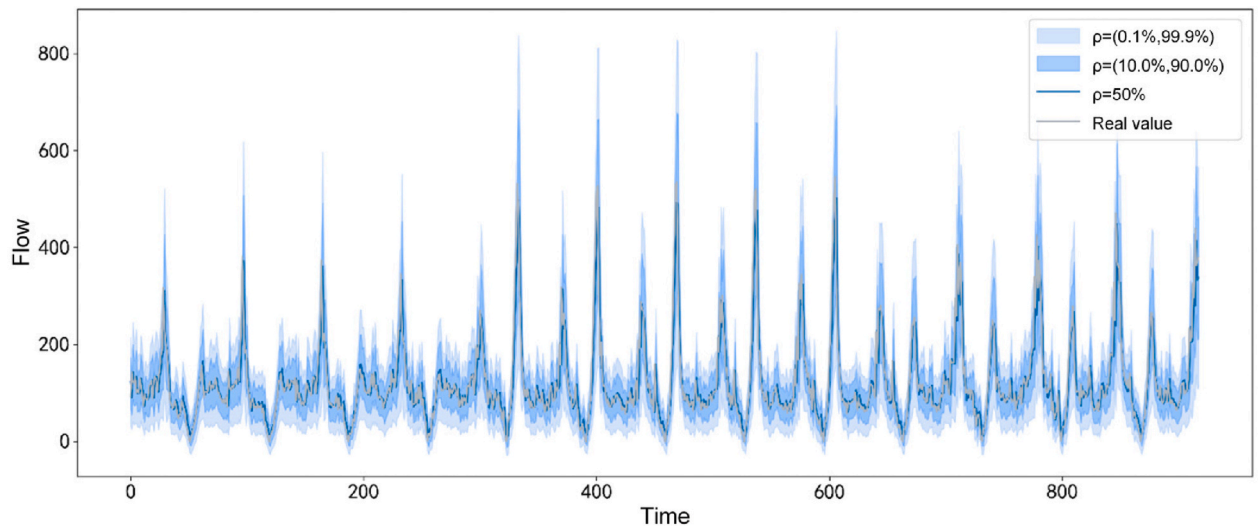


Fig. 16. Passenger flow prediction results in different confidence intervals.

Table 3

The MAE, RMSE and MAPE of different models in the 15-min time granularity.

Model	MAE	RMSE	MAPE/%
DeepAR	19.72	36.11	9.98
N-BEATS	22.82	45.25	10.56
LS-Transformer	25.18	50.82	11.25
TCNN-3	27.85	55.12	12.35
TCNN-4	22.14	43.97	10.35
TCAN	21.01	39.34	10.23
CEEMDAN-TCAN	18.15	33.61	9.81

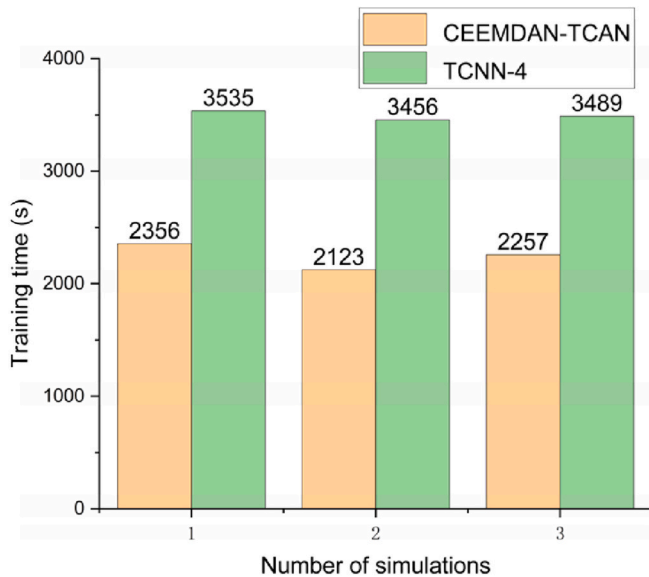


Fig. 17. Training time comparison.

station-level inbound passenger flow forecasting, outperforming other models in both predictive accuracy and training speed, with its probabilistic prediction trends consistently aligning with actual observational patterns. However, a certain prediction lag has been observed, particularly during transitional states between consecutive time intervals—for instance, when inbound passenger volumes exhibit abrupt changes across successive timesteps, the model's outputs demonstrate detectable deviations that typically require one or two subsequent timesteps to

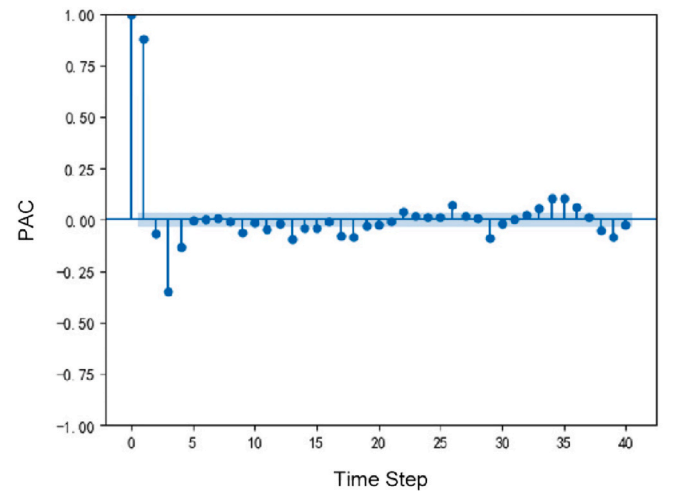


Fig. 18. Partial autocorrelation coefficient test.

properly capture the transition. The partial autocorrelation coefficients reveal statistically significant lag orders (notably lag-2 and lag-5 dependencies at a 40-timestep historical window, as visualized in Fig. 18), which provide prior knowledge for structuring the sparse attention patterns by identifying temporally critical connections between specific historical steps.

The attention weight maps reveal that historical time steps exert significant influence on predictions, and to further validate this rationale, attention score heatmaps at varying time steps were generated alongside verification through partial autocorrelation coefficients (Lin et al., 2021). Fig. 19 presents CEEMDAN-TCAN's prediction results across 30 consecutive time steps in the test set, where the attention diagram displays paired attention scores representing the importance of prior time-series steps (y-axis) in forecasting future steps (x-axis), demonstrating CEEMDAN-TCAN's capability to accurately simulate inbound passenger flow while revealing sparse dependencies between past and future steps, with longer historical sequences proving critical as evidenced by consistently high attention scores for certain early historical steps across all heatmaps; these visualizations elucidate the importance of each input feature for individual instances, enabling both model interpretation and decision justification to enhance trust in practical applications, though comparative analysis of prediction accuracy across different time steps indicates progressive performance

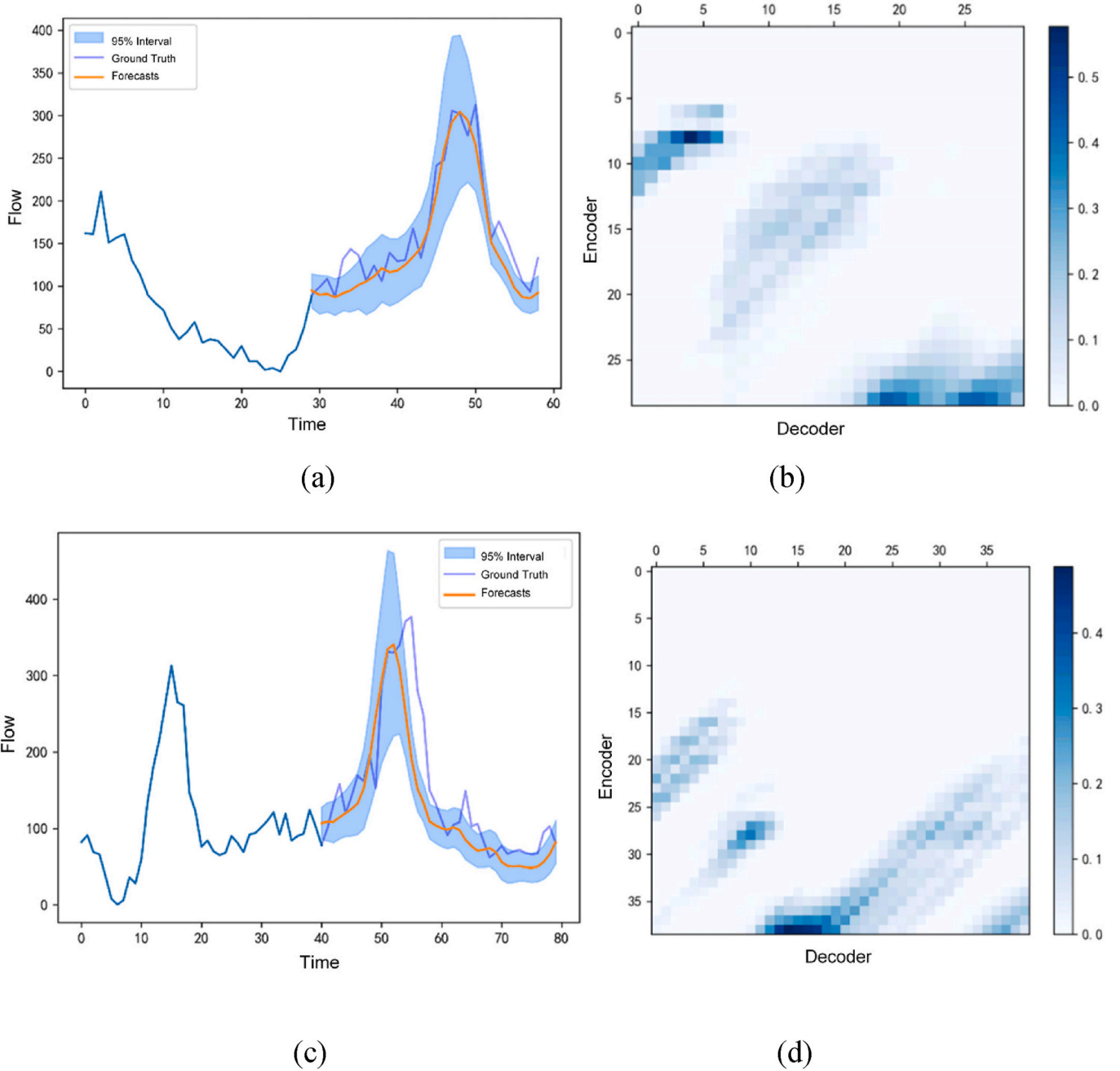


Fig. 19. Prediction accuracy across varying time steps and corresponding attention score heatmaps, (a) the actual values and predicted values under 30 consecutive steps, (b) the attention score heatmap over 30 consecutive steps, (c) the actual values and predicted values under 25 consecutive steps, (d) the attention score heatmap over 25 consecutive steps.

degradation with increasing step lengths due to error accumulation, yet CEEMDAN-TCAN's overall superiority confirms its effectiveness in capturing sparse temporal dependencies and achieving expansive receptive fields without deep architectures while delivering interpretable results through explicit identification of the most consequential time steps for predictions.

5. Conclusion

This study proposes a two-step clustered CEEMDAN-TCAN approach, a composite method for station-level short-term entry passenger flow prediction in urban rail transit systems. First, addressing the characteristics of numerous and heterogeneous stations in rail transit networks, we developed a two-step fuzzy k -means clustering method by

integrating two clustering techniques with rapid convergence and simple principles. This approach performs secondary clustering based on passenger flow attributes and land-use characteristics, excavating spatial features of stations across different parent and sub-clusters, demonstrating better adaptability to temporal distribution patterns and hourly passenger volumes than conventional single-step clustering methods. Subsequently, we employ CEEMDAN for adaptive decomposition of sub-clustered stations to resolve mode mixing issues and reduce reconstruction errors, effectively addressing the automatic adjustment of non-stationary time series data while providing essential input guarantees for entry flow prediction. Furthermore, by incorporating a sparse attention mechanism to enhance TCN, the proposed TCAN method overcomes the limitation of Encoder-Decoder frameworks in comprehensively characterizing full sequence information, ensuring

temporal sequence integrity and influence preservation. Comparative experiments with six classical models demonstrate that CEEMDAN-TCAN achieves 13.61 % MAE reduction, 14.56 % RMSE improvement, and 4.1 % MAPE decrease while requiring 55.8 % less average training time per session, featuring fewer temporal convolutional layers, precise historical step computation, and trainable hyperparameters. Finally, to validate the attention mechanism's rationality, we output attention score heatmaps at varying step lengths and verify them through partial autocorrelation coefficients. The accurate passenger flow prediction not only assists operators and authorities in adjusting station crowd management strategies and optimizing train scheduling, but also enables early warning of potential emergencies for timely passenger flow regulation and peak-hour travel guidance.

However, our model still has many other factors to consider. Due to the privacy and confidentiality of public transportation data, the dataset we obtained only covers a 7-day period, focusing more on standardized operational scenarios for passenger flow prediction, and the feasibility of the proposed model requires further validation. In practice, passenger flow is influenced not only by temporal and spatial factors but also by external disturbances such as weather and holidays, and AFC data cannot fully capture urban residents' travel demands and variations. To highlight the experimental results, we only used partial clustering outcomes to demonstrate the model's superiority on small samples, without in-depth analysis of data fitting. Additionally, the time granularity considered in this study is limited to 15 min, which prevents demonstrating the model's advancement across multiple time scales. These limitations highlight areas needing improvement in terms of adaptability and comprehensiveness.

CRedit authorship contribution statement

Shejun Deng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jiayang Du:** Writing – review & editing, Visualization, Validation, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Jun Zhang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis. **Xiaoying Wang:** Writing – review & editing.

Funding

This work was supported by the National Social Science Fund of China [grant number 23BGL278]; the National Natural Science Foundation of China [grant number 12301081]; Social Science Foundation of Jiangsu Province [grant number 22XZB022], Humanities and Social Sciences Research Planning Foundation of Ministry of Education [grant number 19YJAZH011].

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shejun Deng reports financial support was provided by the National Social Science Fund of China. Jun Zhang reports financial support was provided by the National Natural Science Foundation of China. Shejun Deng reports financial support was provided by Social Science Foundation of Jiangsu Province. Shejun Deng reports financial support was provided by Humanities and Social Sciences Research Planning Foundation of Ministry of Education. Shejun Deng reports a relationship with College of Architectural Science and Engineering, Yangzhou University, Yangzhou that includes: employment. Jun Zhang reports a relationship with College of Architectural Science and Engineering, Yangzhou University, Yangzhou that includes: employment. If there are other authors,

they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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